



Sprint 1 Review: EDA findings and hypotheses presented

Daniel Regalado, Luna Gerlic, Jeanne Hassoun, Lina Graf, Amelia Simpson



Meeting Agenda

1. Problem Statement
2. Data Overview
3. ArcGIS Accessibility EDA
4. ACS U.S. Bureau Census Demographics EDA
5. GTFS Transit System EDA
6. Combined Insights
7. Next Steps & Solution Plan

Problem Statement

Public transit in Miami-Dade suffers from uneven service coverage and limited visibility into demographic access gaps. Planners currently lack predictive tools to evaluate how service changes specifically impact underserved communities.

Key Sub-Questions

Demographics: Which neighborhoods have high "transit dependency" but low access?

Accessibility: What is the "Time Tax"? (Time to jobs via transit vs. car)

Operations: Is service frequency matching potential latent demand?

Business Objectives

Analyze: Identify current equity gaps using fused demographic and transit data.

Predict: Model future demand and identify emerging "transit deserts."

Simulate: Create a "What-If" scenario tool to test service improvements.

Methodology

MVP Approach: Focus on a clean, interpretable pipeline over complex "black box" models.

Data-Driven: Decisions guided by ground-truth metrics

Dataset	Source & Features	Project Purpose
Demographics	US Census Bureau (ACS DP03) - Household income & poverty status - Vehicle ownership & SNAP receipt	The "Who": Identifies vulnerable, transit-dependent populations and maps areas with potential latent demand.
Accessibility	Miami-Dade MPO (ArcGIS LODES) - Job access by time (15 to 60 min windows) - Commute modes (Auto vs. Transit)	The "What": Quantifies the "Time Tax" and measures the spatial equity gap between car owners and transit riders.
Transit System	Miami-Dade GTFS - Route geometries and stops - Trip schedules and frequencies	The "How": Maps the actual physical service to evaluate if bus frequency matches the needs of high-density areas.

What's In The Data

- 36,507 census blocks — all of Miami-Dade County
- 6 transport modes: auto, transit, 4 bicycle stress levels
- 189 columns/block: jobs by wage, age, race, industry
- Employment: LEHD/LODES (Census Bureau, 2019)

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How It's Structured

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- Each row = one census block
- Each cell: "From block X, how many jobs reachable in Y min by mode Z?"
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- Job accessibility: total reachable jobs at each time threshold
- Wage tiers: low ($\leq \$15K/yr$), mid ($\$15-40K$), high ($\$40K+$)
- Mode comparison: auto vs. transit vs. 4 bike safety levels
- Workplace (jobs at destination) + Residence (workers at origin)

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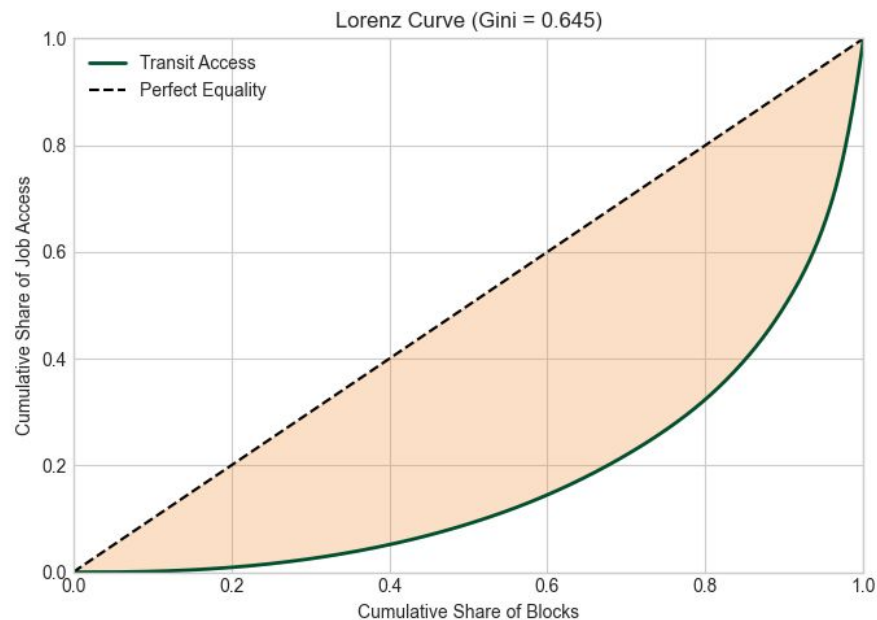
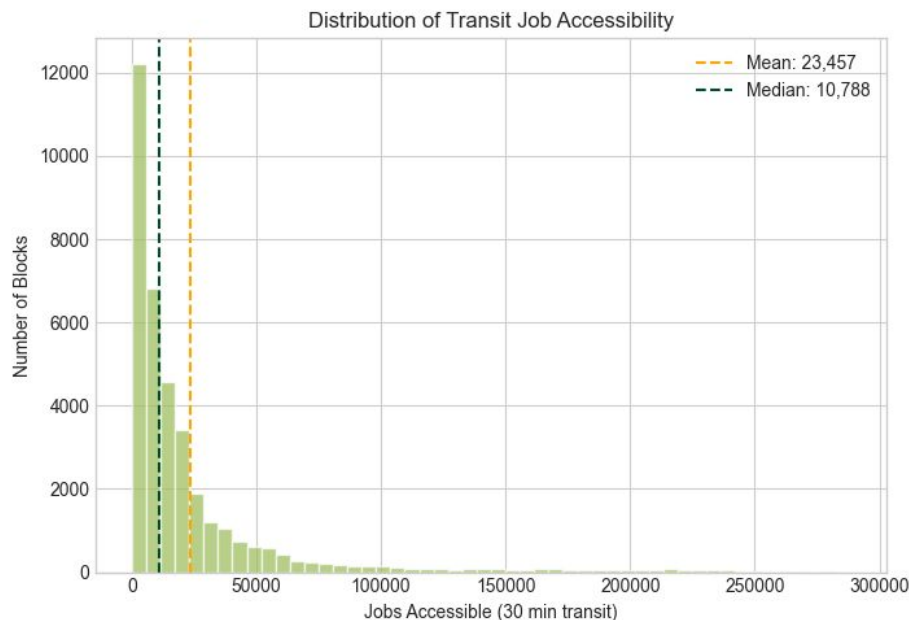
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Key Parameters

- Transit window: weekday 7-9 AM (peak performance)
- Automobile: August 2021 travel times
- Bicycle LTS: 1 = child-safe routes → 4 = any road
- Employment data: 2019 pre-COVID baseline

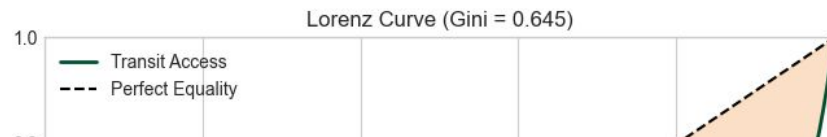
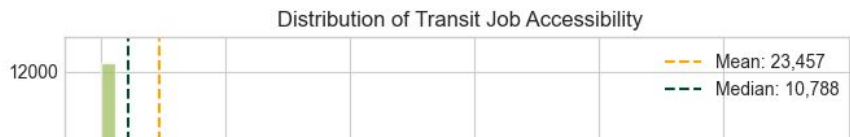
0.645 Gini: Severe Transit Inequality

Higher than US income inequality (~ 0.39). P90/P10 ratio: 64.6x — top 10% of blocks reach 65x more jobs. 751 blocks have zero transit access at 30 min.



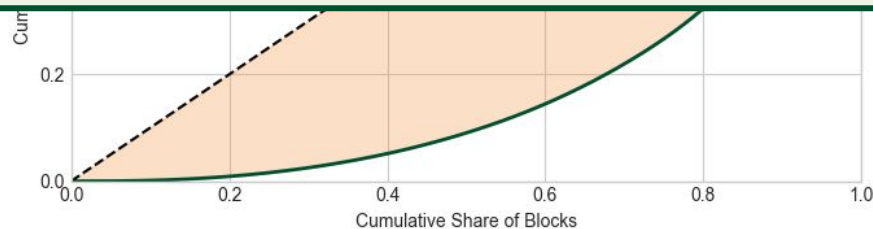
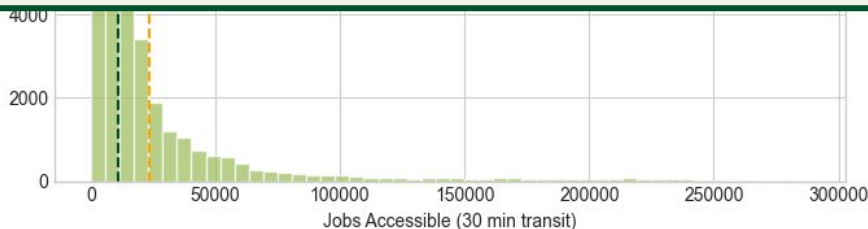
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INSIGHT : Transit follows existing privilege

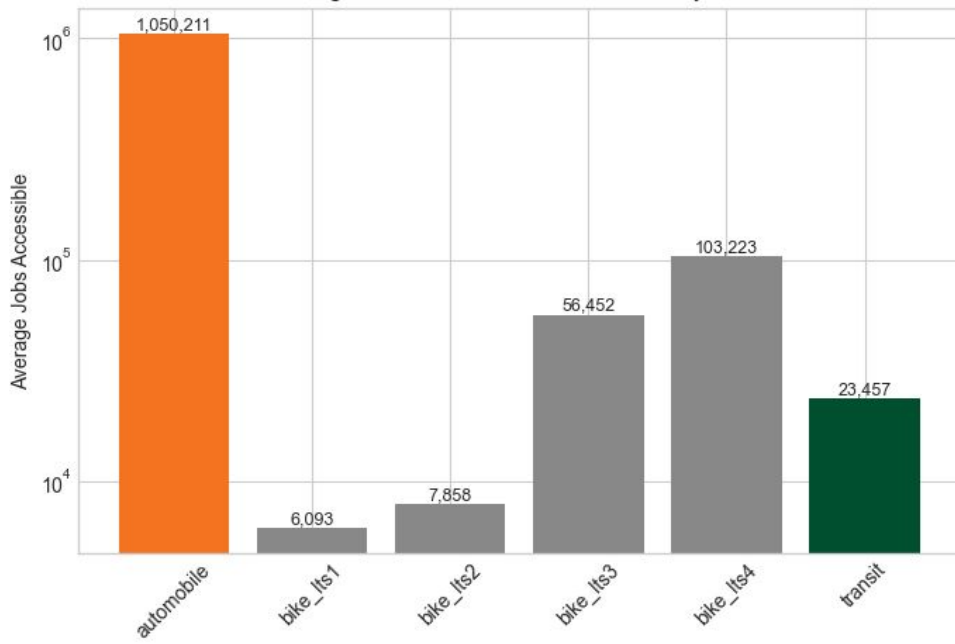
High-auto-access blocks also have high transit access. The current system reinforces geographic advantage. New investment should prioritize coverage gaps, not ridership volume.



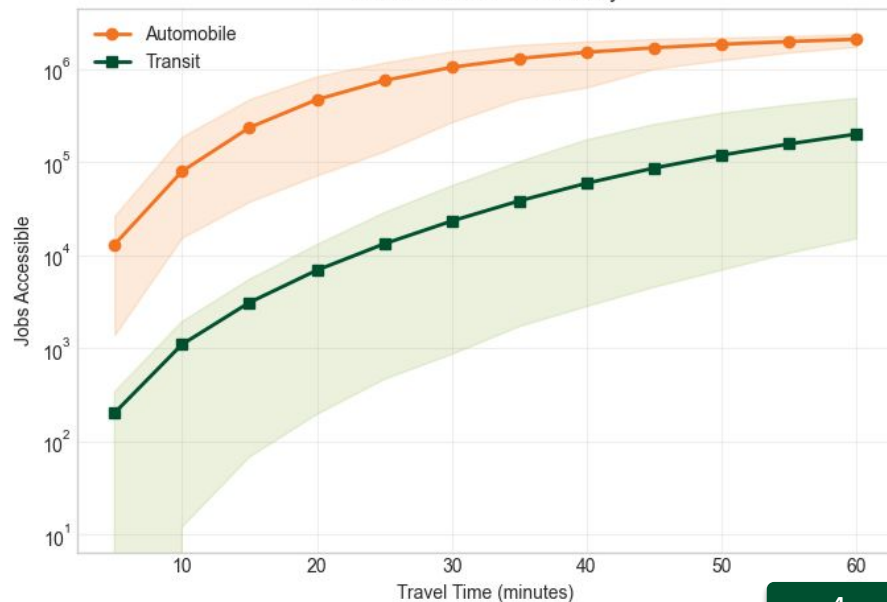
44.8x Auto-Transit Gap

At 30 min: auto reaches 1,050,211 jobs avg vs. transit's 23,457. Worst at short trips (64x at 5 min). Still 10.4x even at 60 min.

Average Jobs Accessible in 30 Minutes by Mode



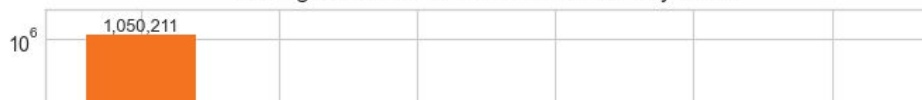
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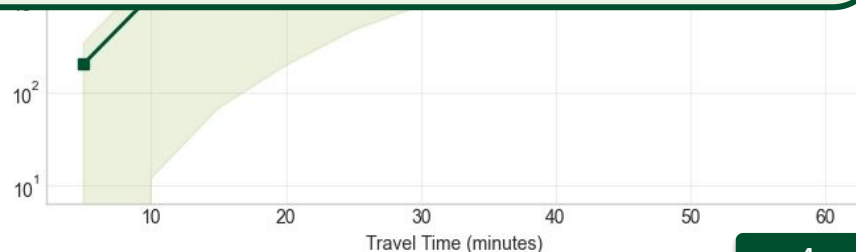
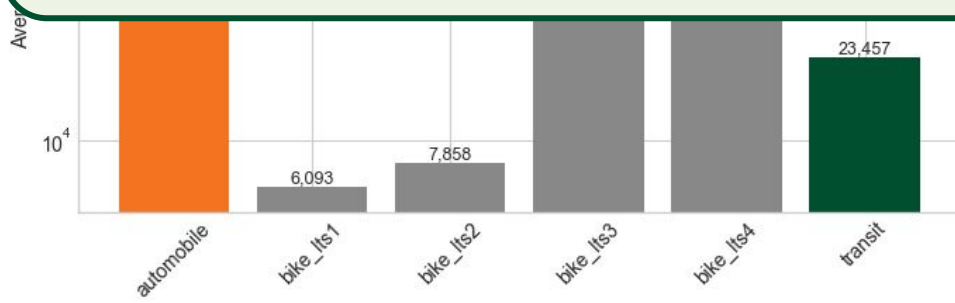


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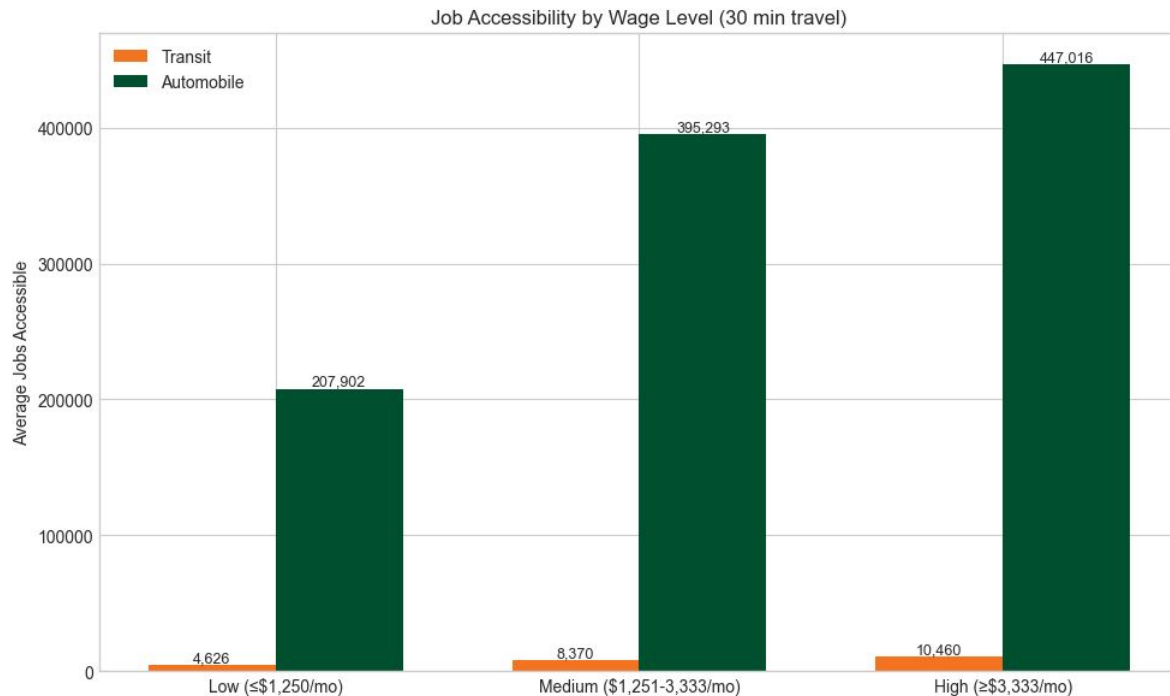
INSIGHT : The gap is structural, not marginal

A 44.8x gap persisting across all time thresholds cannot be closed by optimizing existing bus routes. It requires fundamental mode investment — BRT, rail expansion, or entirely new service patterns.



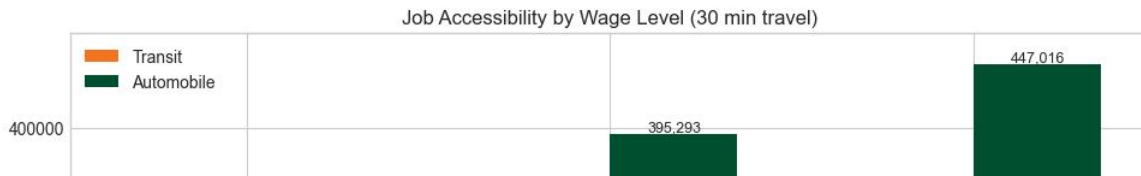
2.26x Low-Wage Workers Face Steepest Barriers

Low-wage jobs accessible by transit: 4,626 avg. High-wage: 10,460. Workers who can least afford cars have fewest transit job options.



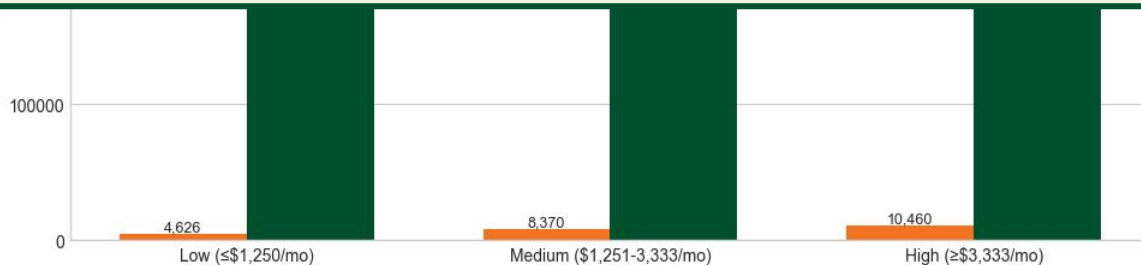
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INSIGHT : Low-wage workers face a triple penalty

Fewer accessible jobs + longer required commutes + higher transportation cost burden as share of income. Transit improvements targeting low-wage job centers would have the highest equity return.

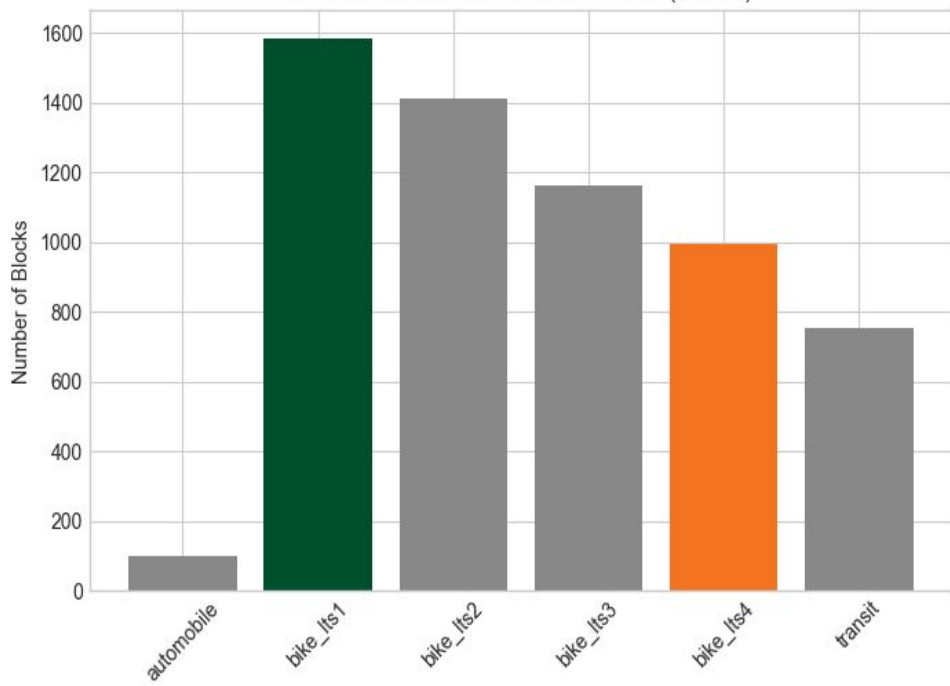


16.9x

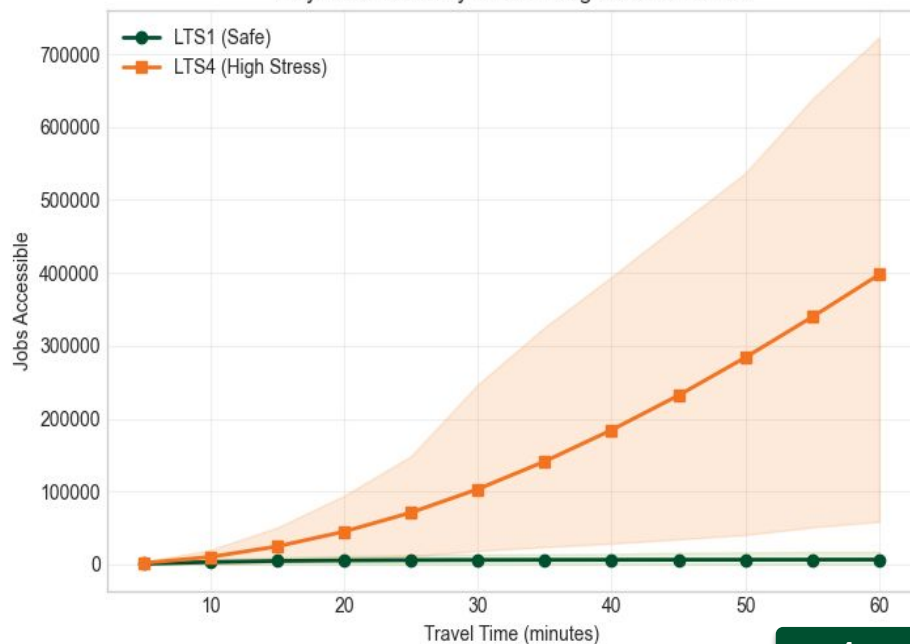
Safe Cycling Is Not Viable

LTS1 (safe routes): 6,093 jobs. LTS4 (any road): 103,223. Safe bike infrastructure is fragmented — not a connected commute network.

Census Blocks with Zero Job Access (30 min)



Bicycle Accessibility: Safe vs High-Stress Routes



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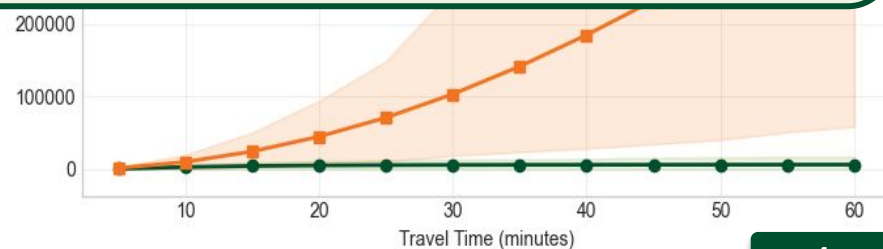
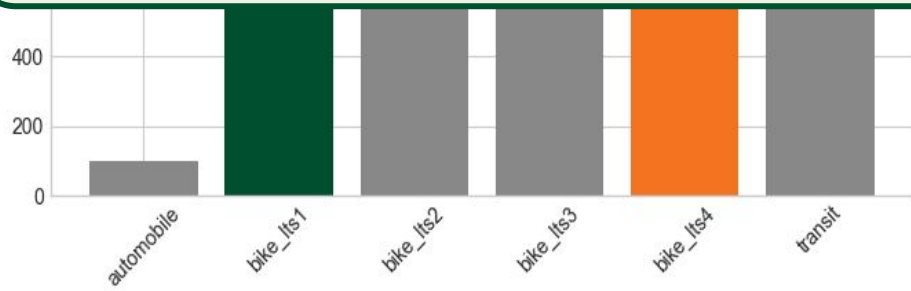


Bicycle Accessibility: Safe vs High-Stress Routes



INSIGHT : 751 zero-access blocks need more than transit

Some blocks are physically isolated — no transit route within 30 minutes reaches any employment. These require land-use and zoning interventions alongside transit investment.



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Its Role In This Project

- Socioeconomic profile of who's affected by access gaps
- Identifies transit-vulnerable populations
- Establishes equity indicators for spatial overlay
- Answers: "What does the population we serve look like?"

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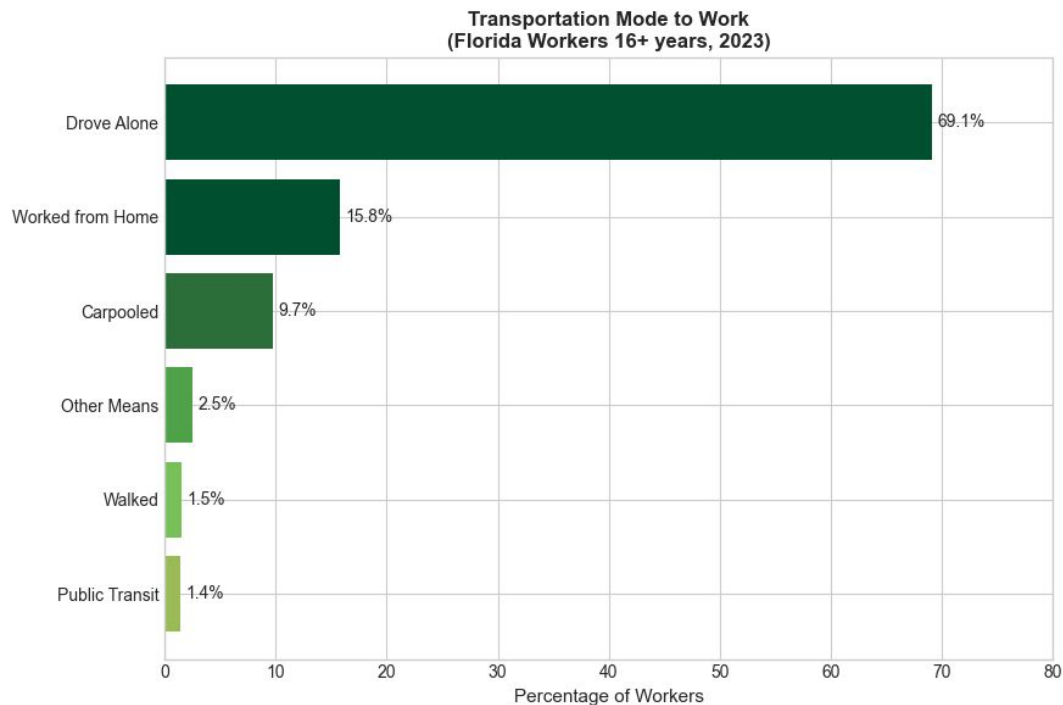
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Scope & How We'll Use It

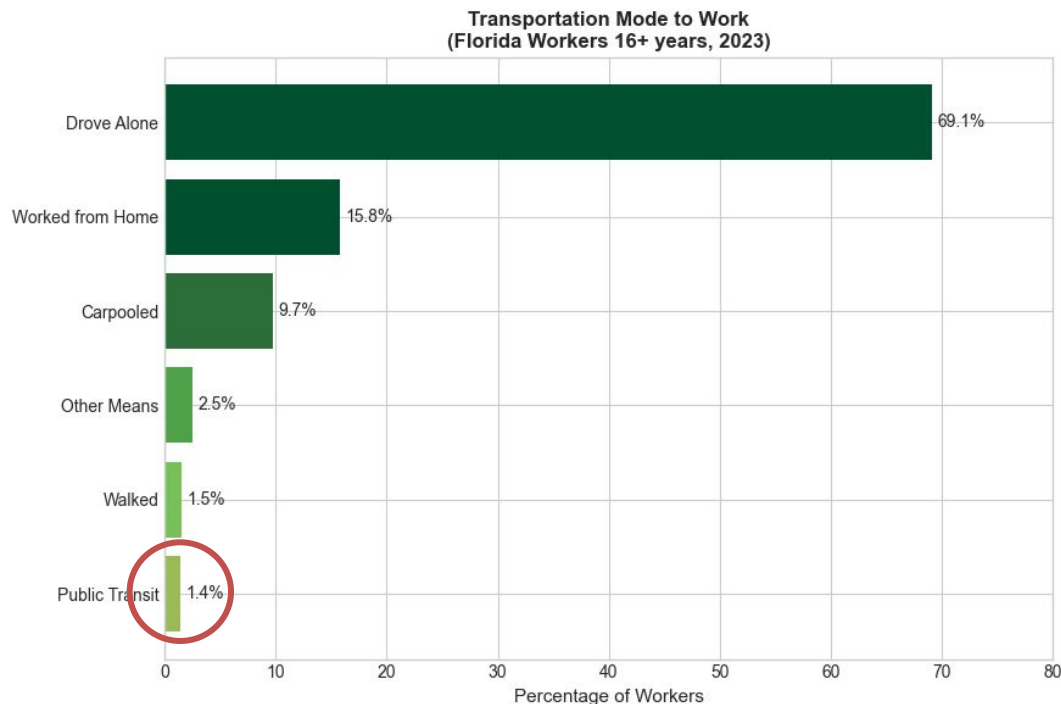
- Florida state-level: economic context + vulnerability baselines
- Miami-Dade targeting: tract-level ACS in Sprint 2
- This sprint: identify WHAT to measure
- Next sprint: locate WHERE those populations live

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Only ~144,000 FL workers commute by transit. 69.1% drive alone. 15.8% work from home (11x transit). Even walking (1.5%) nearly equals transit of 1.4%.

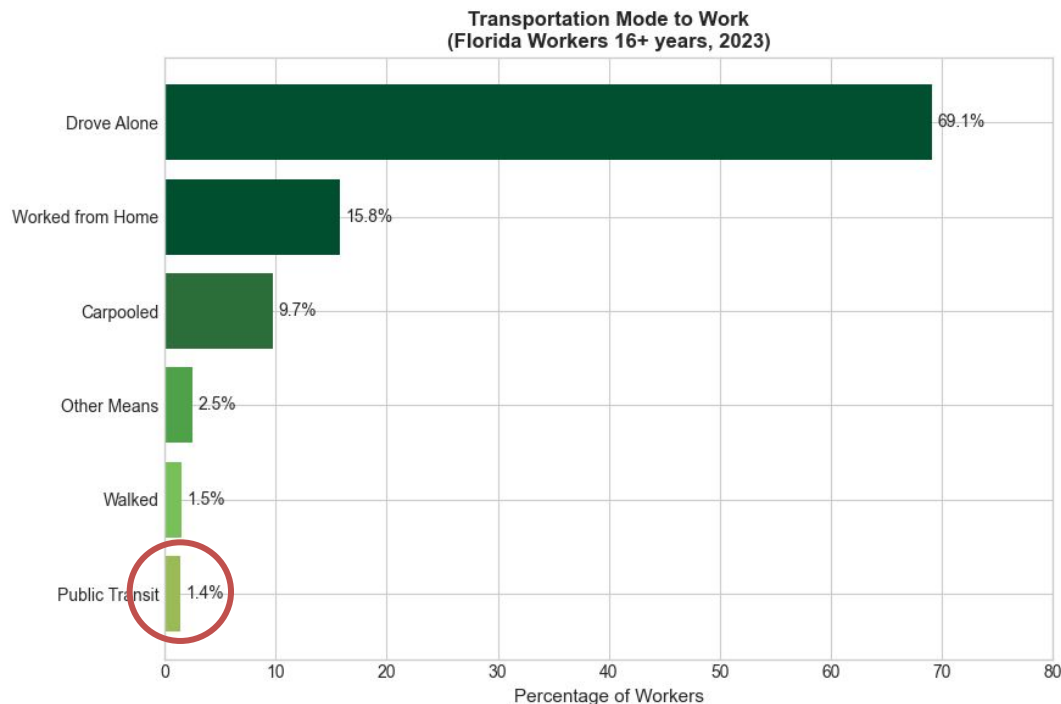


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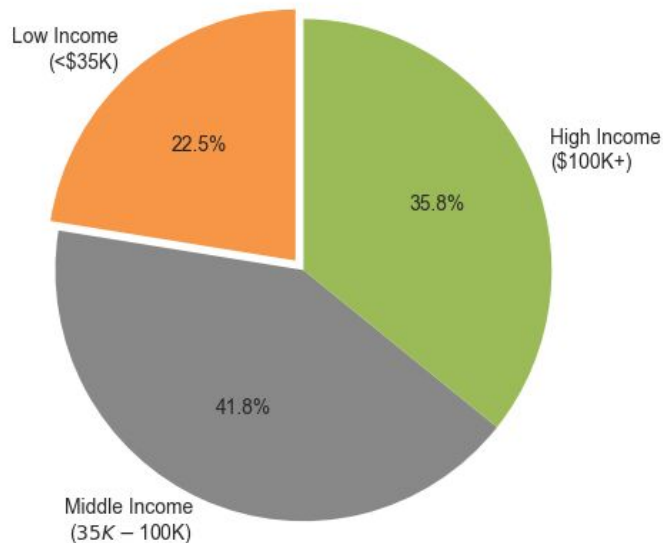
= both the problem and the opportunity

Extremely low usage means the system fails to serve demand. Even a modest shift to 3-4% would double ridership and transform equity outcomes. The question isn't whether demand exists, it's whether the service can capture it.



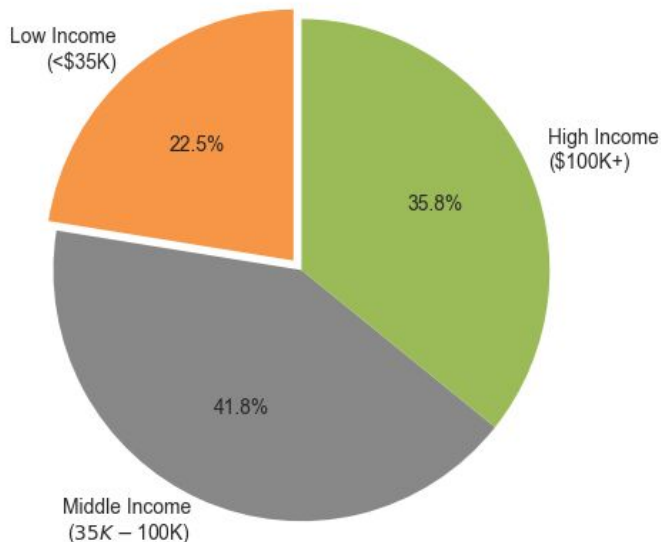
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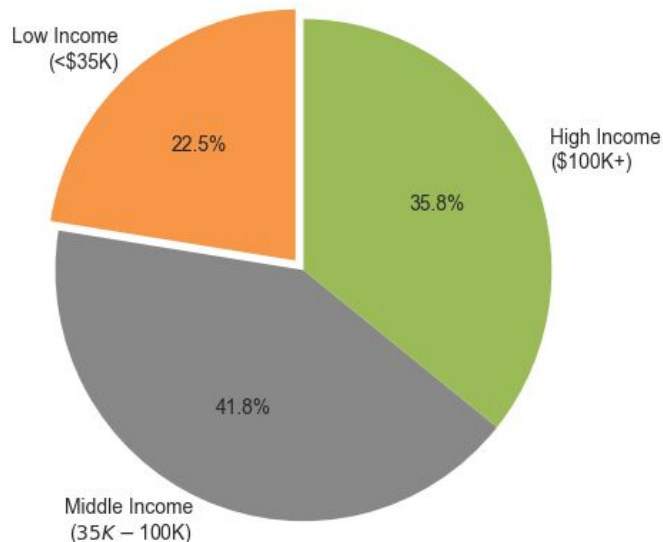
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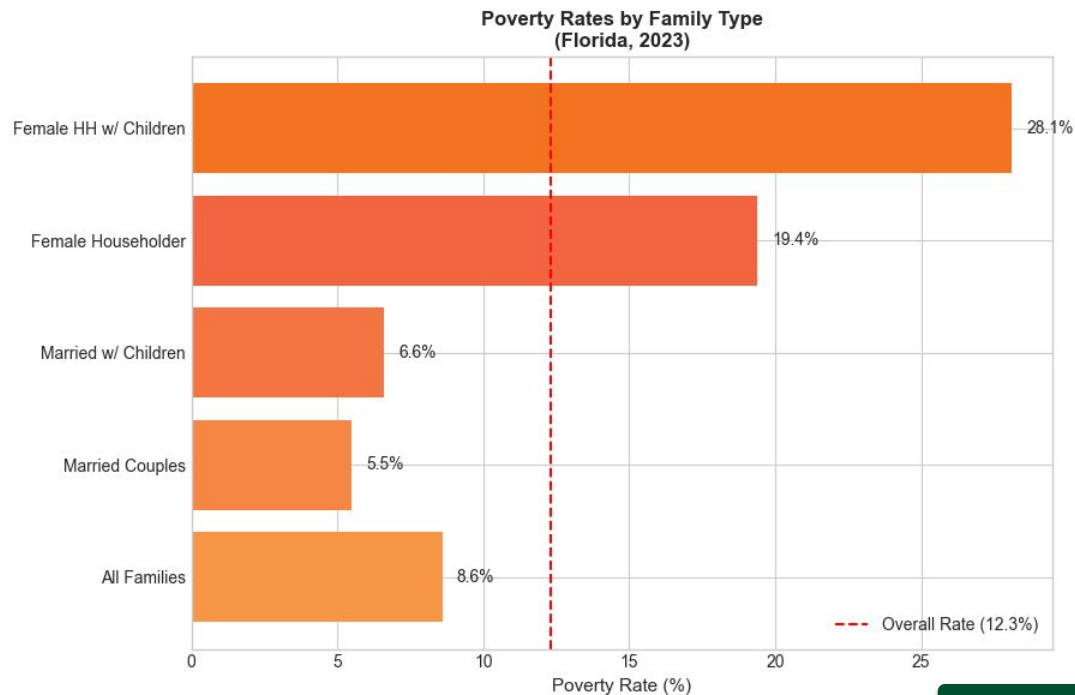


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Job type → income → transit dependency

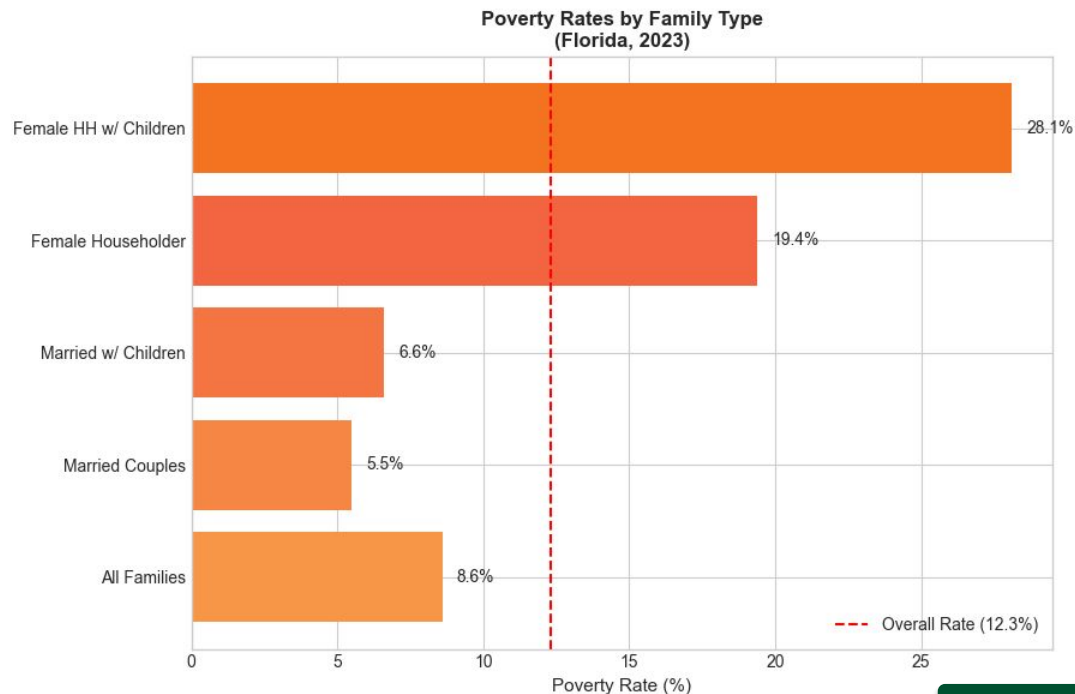
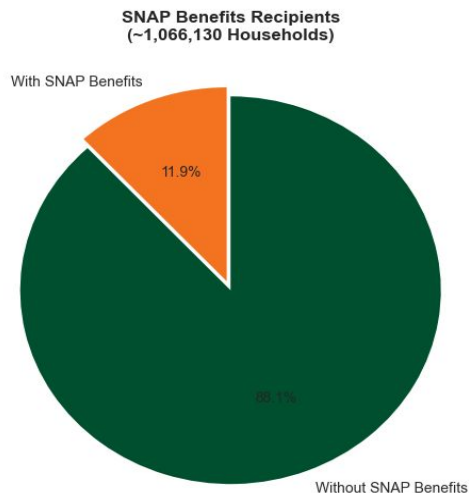
Service and retail jobs pay less. Low income means car ownership consumes 20-30% of earnings — effectively unaffordable. That forces transit dependency on a system that doesn't serve those jobs well. The chain is occupation → income → transportation mode, and it locks low-wage workers into a cycle the current network reinforces.

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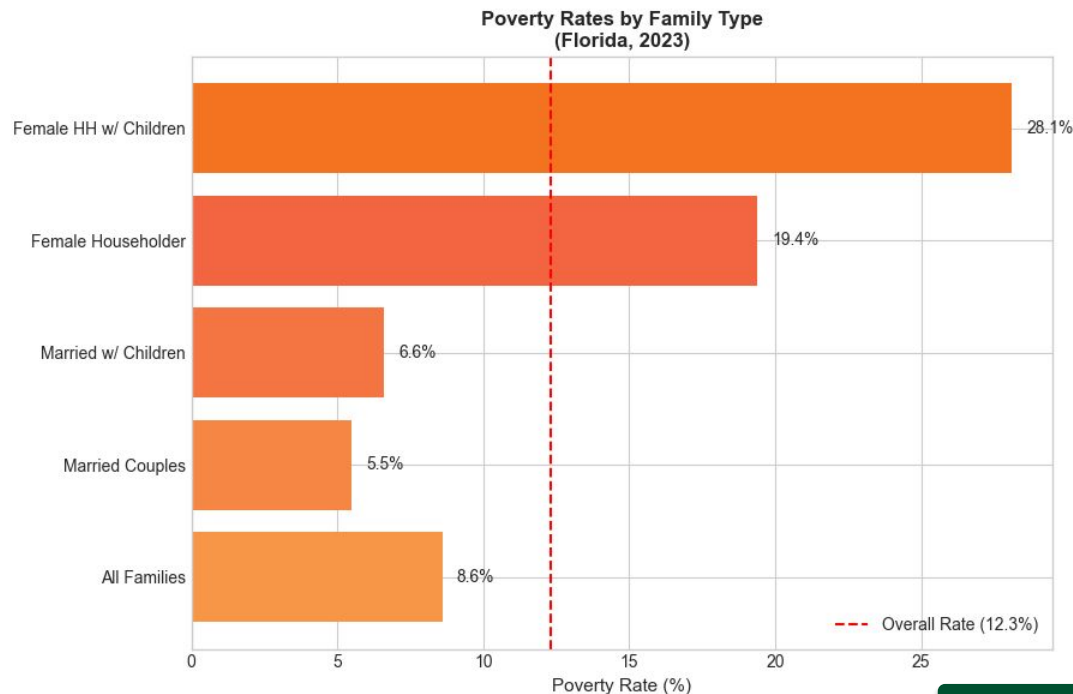


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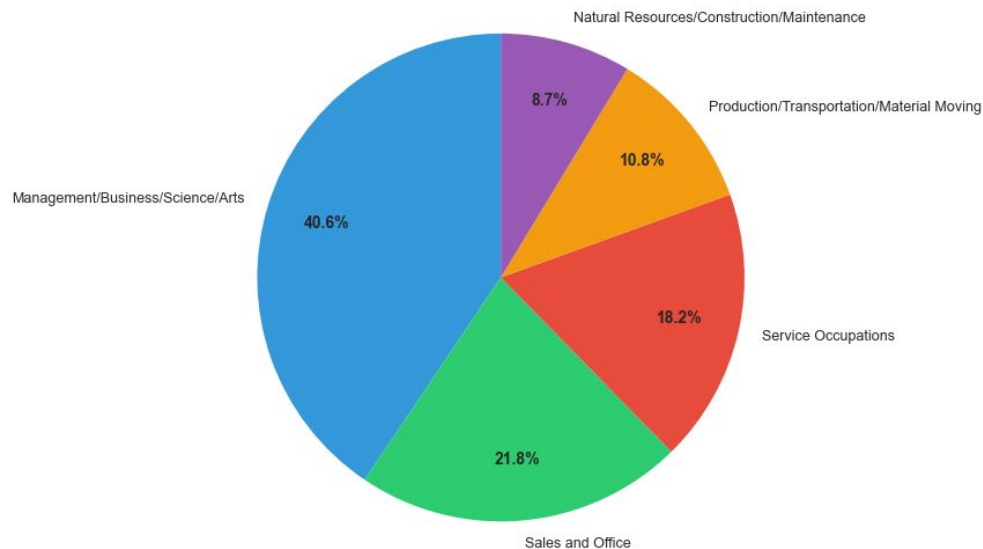
Vulnerability compounds — it's not one variable

Low income + single-parent + service occupation + no insurance creates a transit-captive population with the least flexible schedules and the highest commute burden. Equity targeting must be multi-factor — using income alone as a proxy misses who is actually most affected.



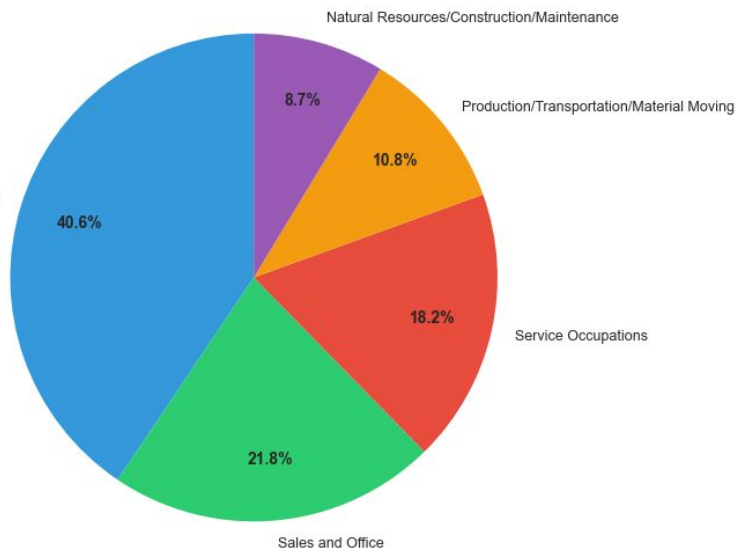
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Occupation Distribution
(Florida Civilian Employed Population, 2023)



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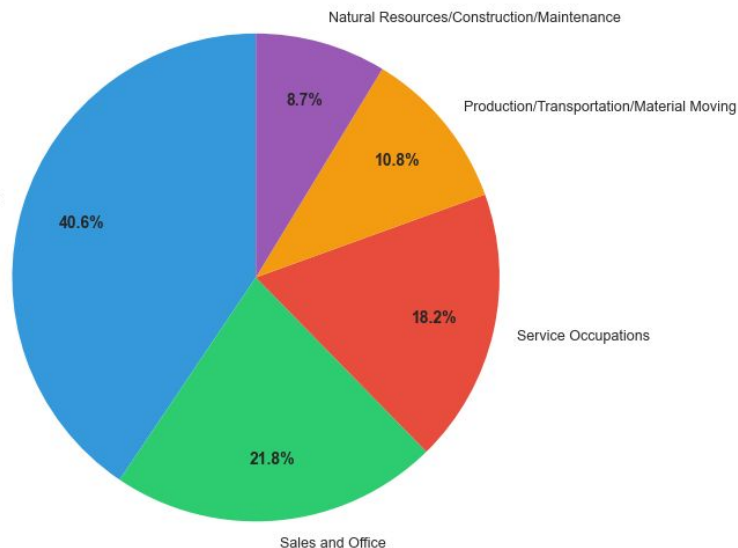
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Shift-based, lower-wage, schedule-variable. These
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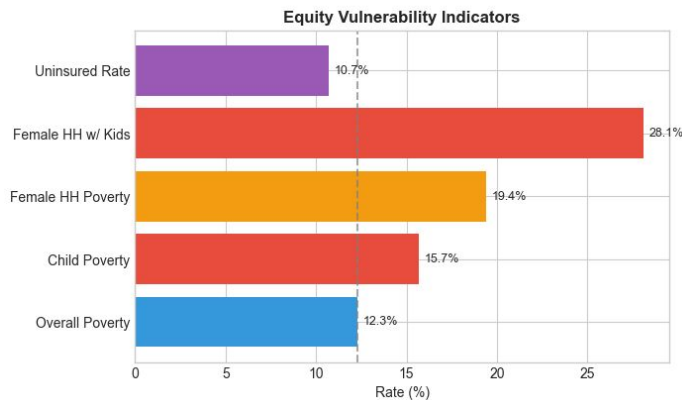
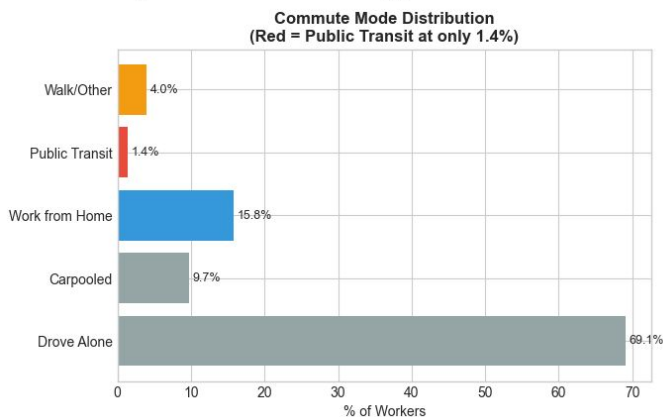
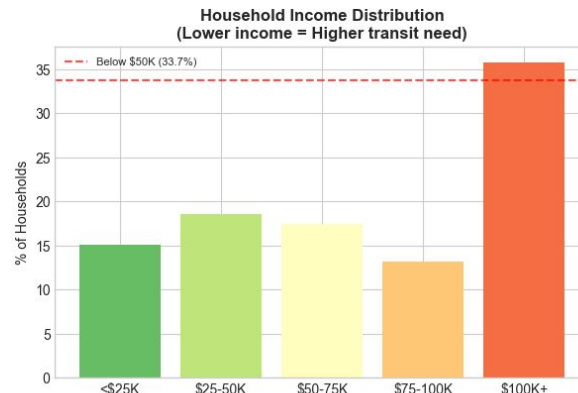
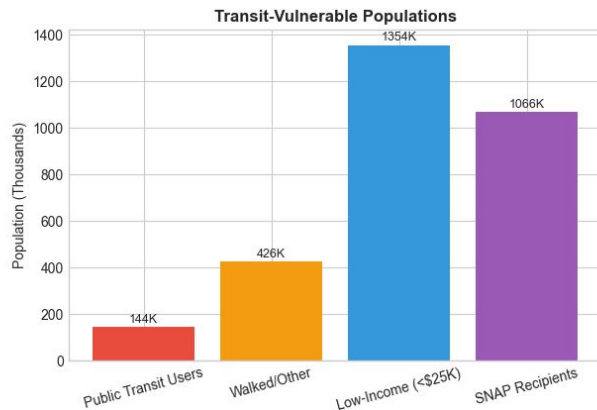


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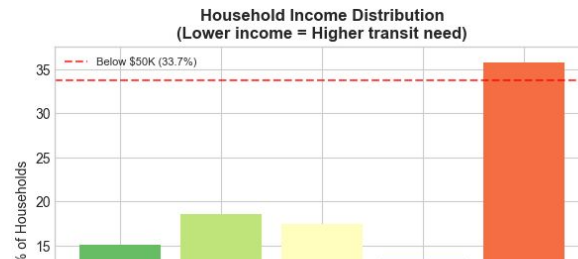
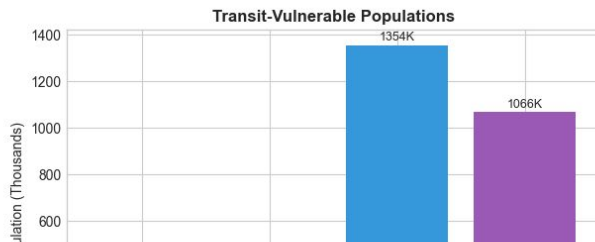
Transit is designed for the wrong workforce

The current system optimizes for 7-9 AM peak commutes — serving the 69% who drive alone. But the 29% who actually need transit work outside that window: opening shifts, closing shifts, weekends, holidays. The mismatch isn't coverage — it's schedule design. **Frequency and span-of-service matter more than route maps for this population.**

Florida Transit Equity Dashboard - ACS 2023 Data

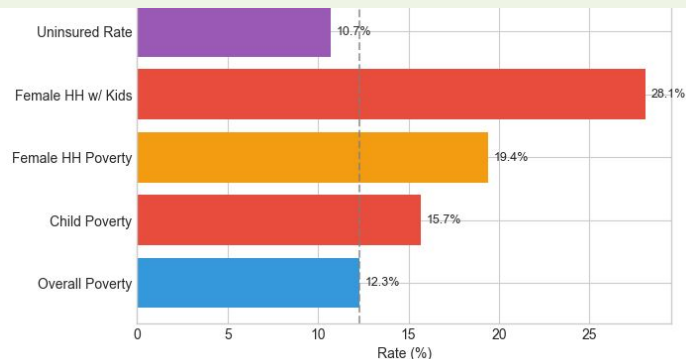
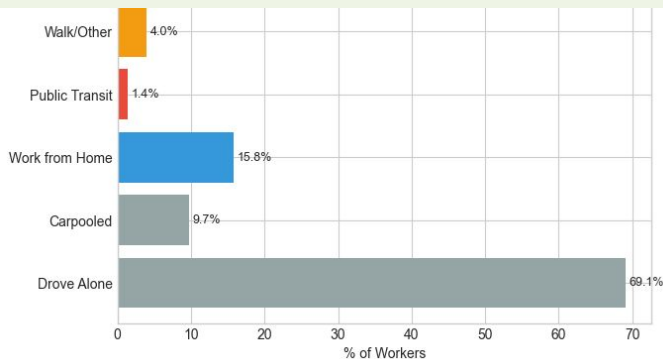


Florida Transit Equity Dashboard - ACS 2023 Data



This EDA identified WHAT to measure.

Sprint 2 applies these indicators WHERE they matter — at Miami-Dade block level, overlaid with ArcGIS.



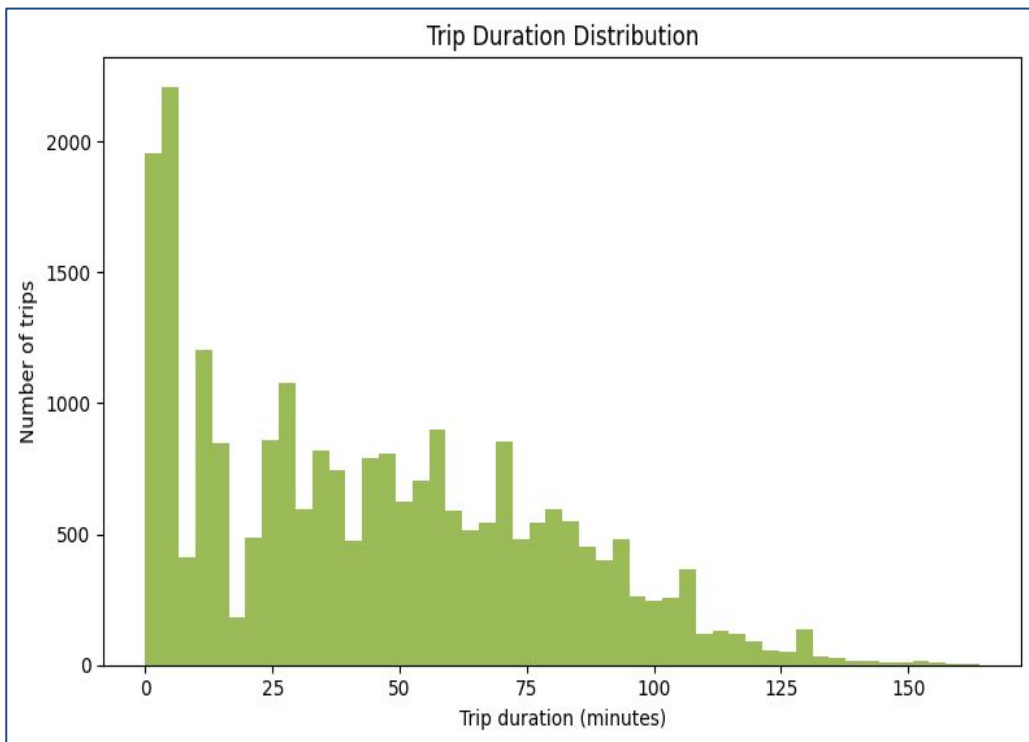
WE HAVE	WE STILL NEED
State-level vulnerability indicators	Tract-level ACS for Miami-Dade neighborhoods
Income and poverty baselines	Vehicle availability per household
Commute mode shares	Commute flow data (origin-destination)
2023 estimates (most current ACS)	Post-COVID employment location shifts

Quick stats found:

- Typical trip durations are long with a mean of **45.58** minutes and they average **~38.46** stops, indicating they are long and stop-dense

Top routes include:

1. MIA (Metrorail, Metrobus, Metromover)
 2. MMO/MMI (Brickell Metromover)
- Bus is the most popular transit system with over 120 routes, followed by streetcar and rail
 - More trips are outbound (away from city center) than inbound

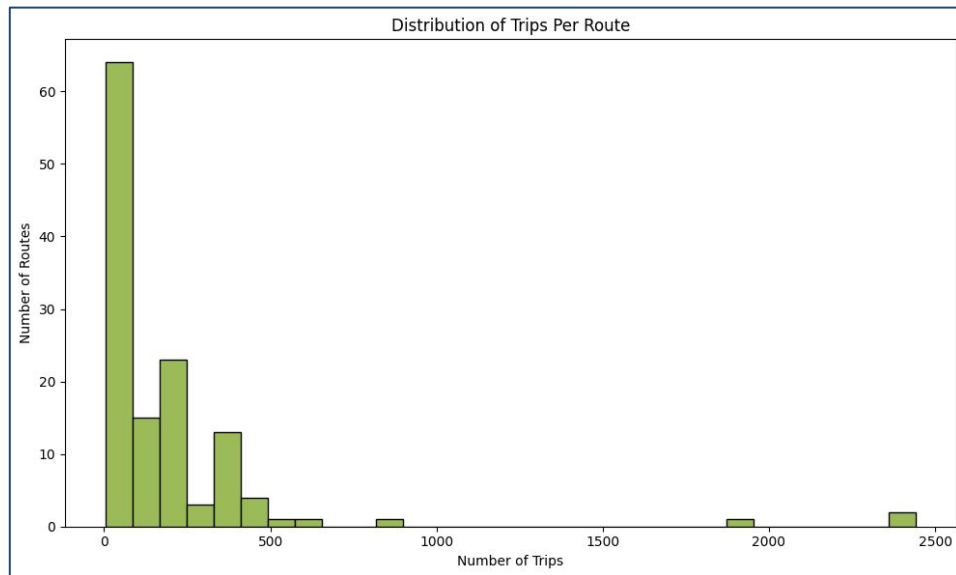


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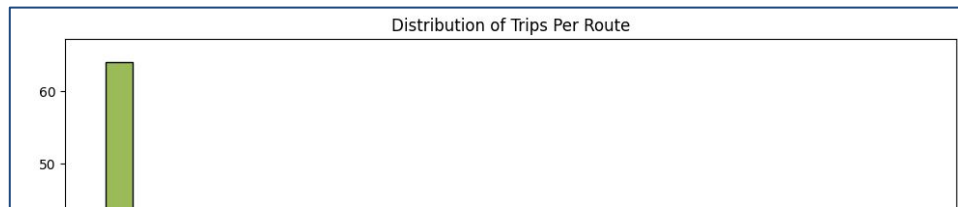
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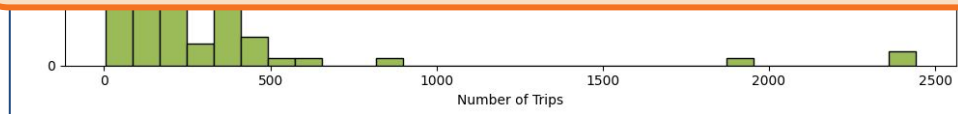
Trip Distribution Equality:

- A Gini coefficient of **0.6330** = relatively high
- Indicates a substantial level of inequality in the distribution of trips across routes
- A value close to 0 would suggest more equal distribution, while a value closer to 1 would indicate maximal inequality.
- The calculated value suggests that a significant portion of the total trips are concentrated on a limited number of routes.



INSIGHT :

Highly skewed distribution, with a large number of routes having a relatively low number of trips and few routes are significantly busier than the majority.



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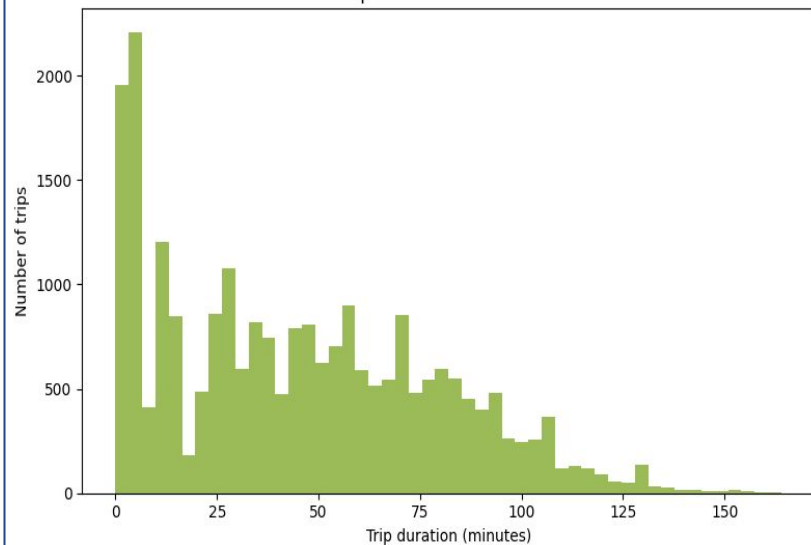
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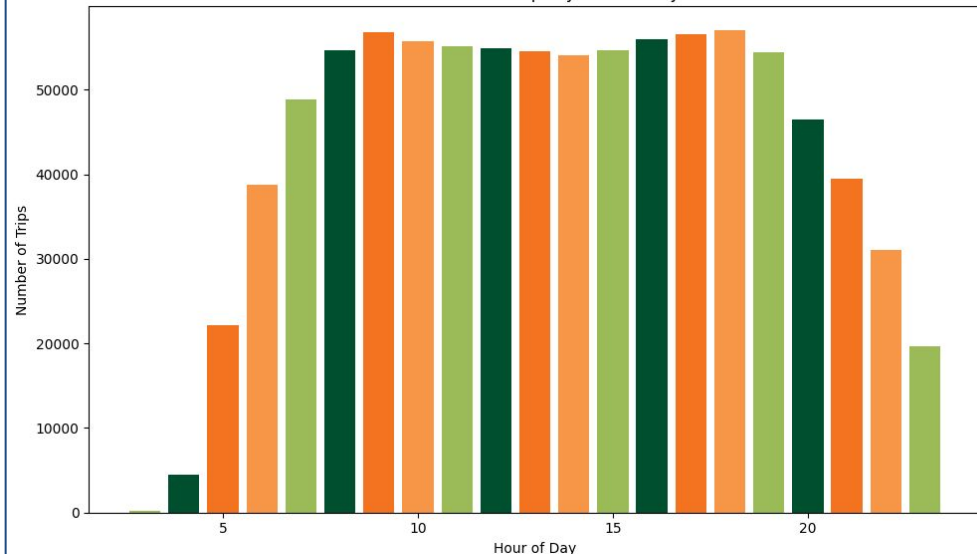
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- **6529 stops** were marked as 'Not Accessible'
- There were **0 stops** with 'No Info' regarding wheelchair boarding, indicating that all stops had a defined status



Trip Duration Distribution



Distribution of Trips by Hour of Day



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It is schedule data, not ridership data

This is GTFS schedule data (routes, trips, stop_times, calendar, shapes). It tells us:

- What *should* run
- When it's scheduled
- Where stops are located

It does **not** tell us:

- How many people ride each trip
- Which stops are overcrowded
- Which neighborhoods are underserved
- Which low-income areas lack stop density

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How it connects to ArcGIS and the Census

Census is the “Demand Layer”, ArcGIS is the “Accessibility Layer”, and GTFS is “Network Supply Layer”

- **ArcGIS:** gives info that a certain block has access to 23,000 jobs in 30 minutes by transit
- **GTFS:** *which* specific bus routes and rail lines make that access possible
- If we want to improve access, GTFS is where we model the intervention such as adding a route, increase frequency, or extend hours

Dataset Limitations

It is schedule data, not ridership data

This is GTFS schedule data (routes, trips,

The GTFS data gives us the operational reality behind the ArcGIS accessibility numbers — these are the actual routes and schedules that produce the gaps we measured.

- How many people ride each trip
- Which stops are overcrowded
- Which neighborhoods are underserved
- Which low-income areas lack stop density

How it connects to ArcGIS and the Census

Census is the “Demand Layer”, ArcGIS is the “Accessibility Layer”, and GTFS is “Network Supply Layer”

to measure and understand even as changing a route, increase frequency, or extend hours

Three Datasets, One Equity Framework

Each dataset answers a different question. Together they answer THE question:

Which vulnerable populations live in areas with the worst transit access?

WHERE | ArcGIS

Spatial accessibility data

- 751 blocks with zero transit access
- Gini coefficient: 0.645
- Low-wage jobs 2.26x less accessible by transit
- Auto-transit gap: 44.8x



WHO | ACS Census

Demographic vulnerability

- 22.5% low-income households
- 28.1% poverty rate for female HH with children
- 29% shift workers needing off-peak transit service



HOW | GTFS Transit

Operational service data

- Routes, frequencies, and schedules that produce these gaps
- Trip distribution
- Gini: 0.633
- Service supply per block

= Priority Target: Blocks where geographic isolation meets economic vulnerability, served by inadequate routes

Key Equity Metrics Identified

From our EDA, we identified the features and KPIs that will drive modeling in Sprint 2:

Accessibility Features (from ArcGIS)

- Jobs reachable by transit at 30 min (total, low-wage, high-wage)
- Auto-Transit Gap Ratio per block (avg 44.8x)
- Wage-tier accessibility disparity (low vs. high-wage job reach)
- Zero-access block flag (binary: 751 blocks identified)

Vulnerability Indicators (from ACS Census)

- Household income bracket (<\$35K threshold)
- SNAP receipt rate (proxy for concentrated poverty)
- Female-headed household poverty rate
- Service/production worker concentration

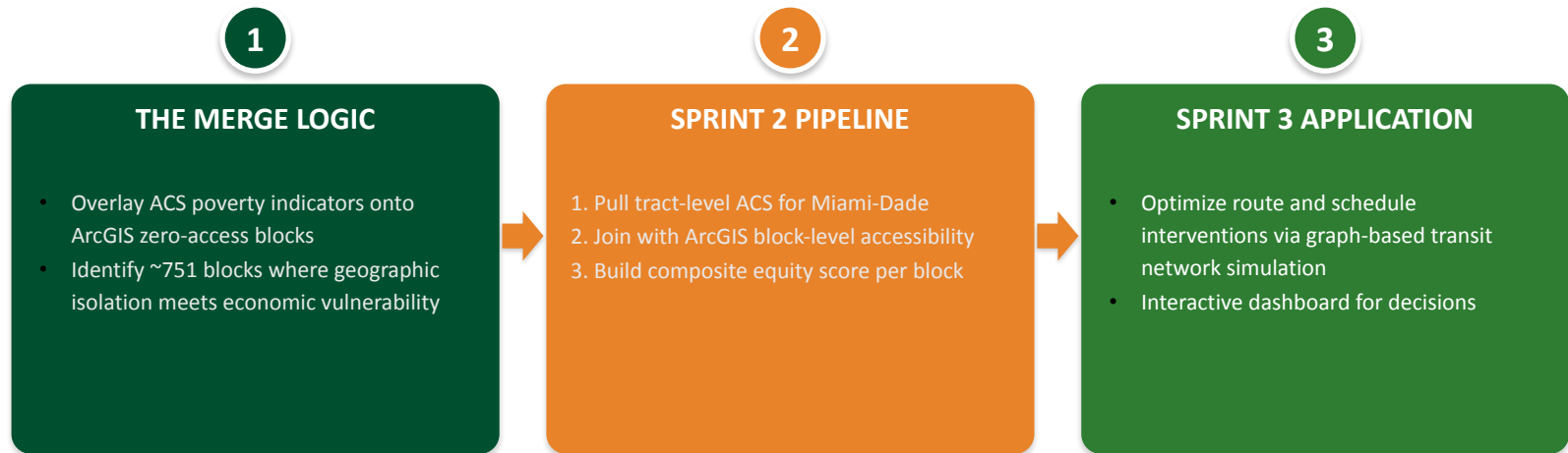
Service Supply Metrics (from GTFS)

- Route frequency and span-of-service per block
- Stop density and network connectivity
- Trip distribution inequality (Gini 0.633)

Target Variable: Composite Equity Score per census block combining accessibility gap + vulnerability indicators

From EDA to Model: The Overlay

How Sprint 1 findings feed into Sprint 2 modeling and Sprint 3 application



Sprint 1 identified WHAT to measure and WHERE the gaps are.
Sprint 2 will model WHO is most affected and PREDICT where gaps will grow.

Quality Assessment

Dataset	Completeness	Reliability	Currency
ArcGIS	Full coverage 36,507 blocks, no gaps	High — LEHD/LODES (federal source)	Employment: 2019 Network: Aug 2021
ACS	91.1% complete MOE ±19,685 avg	High — Census Bureau 90% confidence	2023 1-year estimates
GTFS	Scheduled service only No real-time or ridership	Standard — official agency feed	Current feed Miami-Dade Transit

GTFS-Specific Limitations

- Schedule ≠ Reality — planned times, no delay/cancellation data
- No ridership counts — cannot measure actual demand vs. capacity
- No real-time feeds (GTFS-RT) — static schedule snapshot only
- Route geometry without street-level transfer walking paths
- Fare structure not included — cost burden not calculable from feed

Key Assumptions & Caveats

Pre-COVID employment baseline

ArcGIS uses 2019 jobs data. Low-wage/service jobs (our equity focus) are less affected by remote-work shifts — treated as stable baseline.

Peak-only transit measurement

Transit measured 7–9 AM weekday only — transit at its best. Off-peak and weekend gaps would be larger. Our numbers are conservative.

State vs. county scope (ACS)

ACS DP03 is Florida-level — used to identify which indicators predict vulnerability, not for geographic targeting. Tract-level resolves this in Sprint 2.

Scheduled ≠ actual service (GTFS)

GTFS reflects planned schedules, not real-world performance. Delays, cancellations, and crowding are not captured. Validation against GTFS-RT planned.

Access ≠ usage

We measure what people CAN reach, not what they DO use. Ridership data complements this in later sprints.

Static network snapshot

Transit network captured at a point in time. Route changes since not reflected. Will validate against current GTFS.

Acknowledging limitations upfront strengthens credibility — every caveat has a mitigation plan in Sprint 2.

Moving on to modeling

Our EDA flags a large population who are not being served by the current transit network: women, service workers and low-income households.

Route data shows that low-income blocks were isolated from access to jobs compared to higher-income areas.

Therefore, we believe we can use the next modeling stage to identify **transit deserts**. The people who are underserved by the transit network do not appear in transit data because they **do not have access**. Modeling will identify these areas and lead us to potential interventions.

MVP

Level 1:

Transit model with the power to identify service gaps, predict future demand (**machine learning**), the inputs of which will feed into an optimization model to optimize service and bridge access inequity gaps.

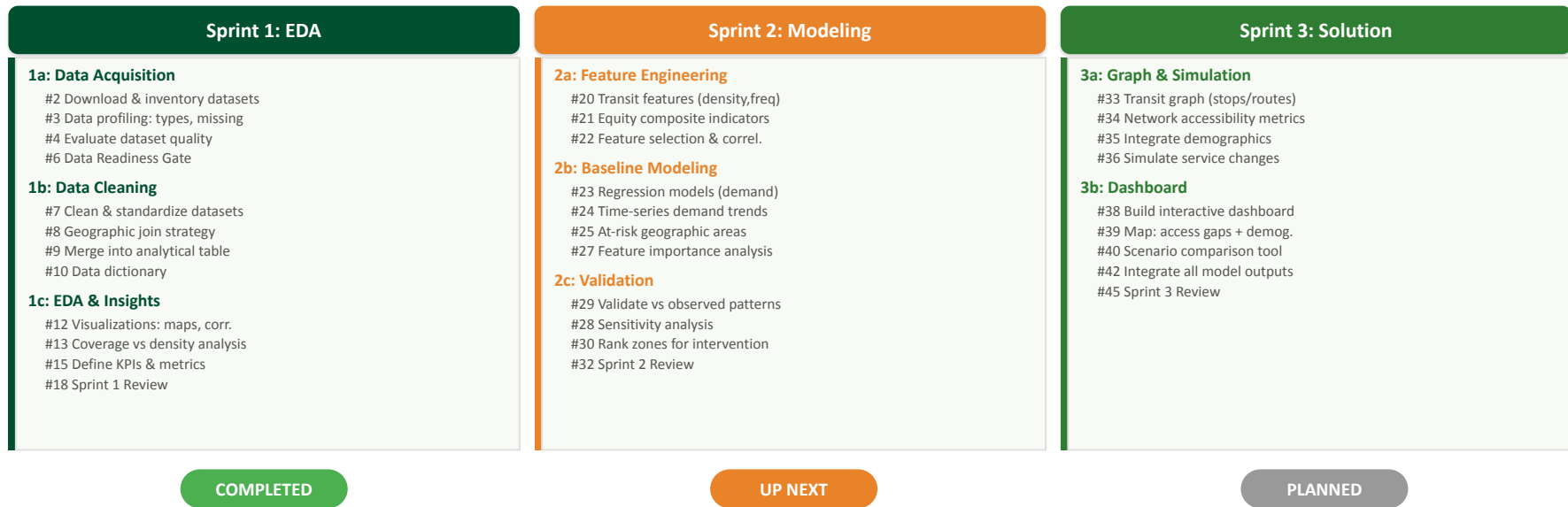
Level 2:

Simulation of proposed solutions to our previously identified service gaps

Level 3:

Interactive dashboard that maps access gaps, visualizes proposed network changes and resulting equity indicator impacts

Sprint Roadmap



Project Resources & Links

All deliverables, code, and documentation for the capstone project

GitHub Repository

55 issues (tasks) across 5 sprints

Notebooks, data, models

github.com/DanielRegaladoUMiami/CapstoneProject-AI-for-Equitable-Transportation

Data Sources

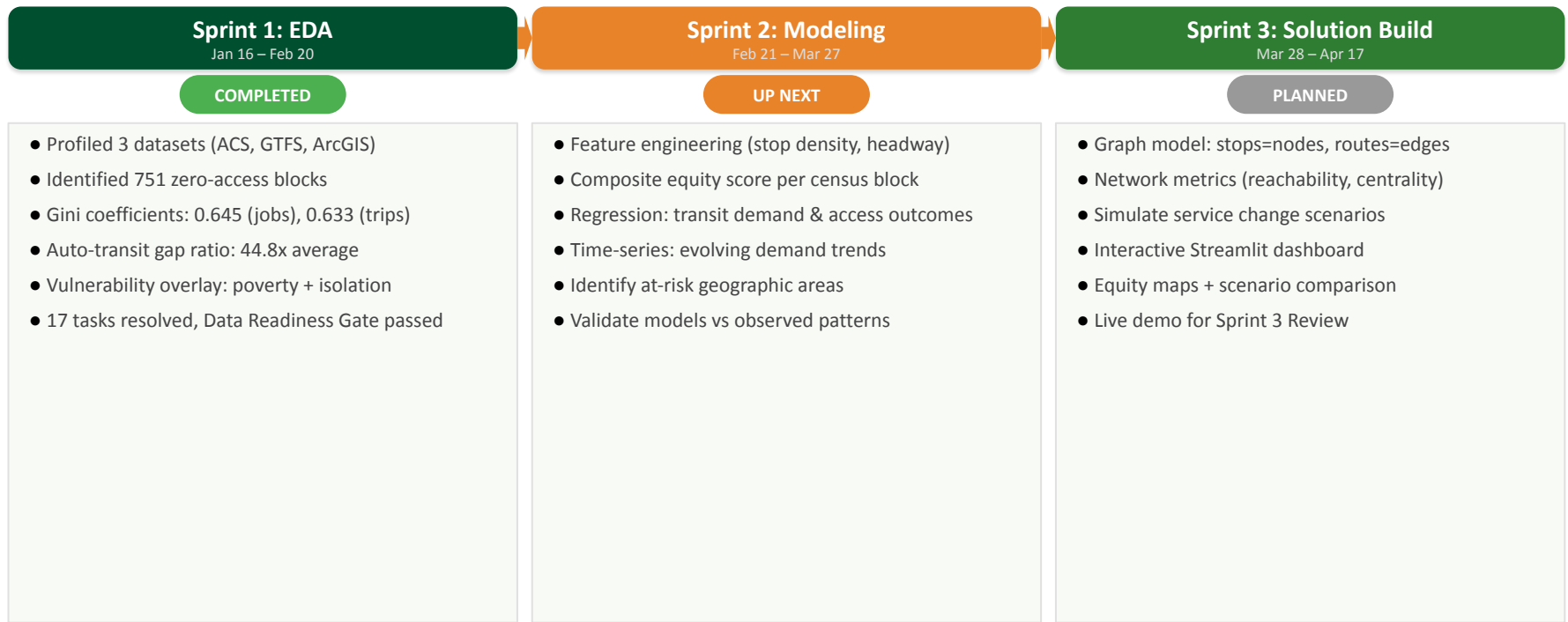
ACS Census • ArcGIS Accessibility

GTFS Transit • Miami-Dade MDT

[See Data/ folder in repository](#)

55 Issues • 5 Sprints • 8 Sub-sprints • 3 Decision Gates • Managed via GitHub Projects

What's Next



Goal: Build an AI-powered model to identify transit deserts and simulate equitable interventions for Miami-Dade County



Q&A